Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

ANS:

Here is a table outlining 5 common network threats paired with real-world examples from popular apps and services:

Network Threat	**Description**	**Real-World Example in Popular Apps**
Phishing	Fraudulent attempts to steal sensitive user information by masquerading as a trusted entity.	Receiving a fake SMS or WhatsApp message claiming to be from **Paytm** or **Flipkart** stating your account is blocked, prompting you to click a malicious link to enter your credentials.
DDoS (Distributed Denial of Service)	Overwhelming a target server with traffic to render its services unavailable to genuine users.	Attackers flooding **Instagram** or **WhatsApp** servers with high volumes of automated requests, causing temporary global outages where users cannot send messages or refresh feeds.
Man-in-the-Middle (MitM)	An attacker secretly intercepts and relays communications between two parties who believe they are directly communicating.	A user connecting to an unsecured public Wi-Fi network at a café, allowing a hacker on the same network to intercept unencrypted traffic or insecure API calls made inside an app.

Malware	Malicious software designed to infiltrate, damage, or steal data from a device or application ecosystem.	A user downloading a modified, unofficial version of **WhatsApp** (like WhatsApp GB) from a third-party website containing spyware that steals personal chats and logs keystrokes.
Password Attacks	Attempts to gain unauthorized access by guessing passwords through brute-force, credential stuffing, or dictionary lists.	Attackers using leaked credential databases from other data breaches to automate login attempts on **Instagram** accounts that use recycled passwords.

Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.

ANS:

```
import time

def simulate_brute_force(target_pin):
    print(f"Target PIN: {target_pin}")
    print("Initiating brute-force attack...\n")

    attempts = 0
    start_time = time.time()

    # Iterate through all 4-digit combinations (0000 to 9999)
    for i in range(10000):
        # Format the number as a 4-digit string with leading zeros
        guess = f"{i:04d}"
        attempts += 1

        # Check if the guess matches the target PIN
        if guess == target_pin:
            elapsed_time = time.time() - start_time
            print(f"✅ PIN Found: {guess}")
            print(f"📊 Total Attempts: {attempts}")
```

```
            print(f"⏱️ Time Taken: {elapsed_time:.4f} seconds")
            return attempts

    print("❌ PIN not found.")
    return 0

if __name__ == "__main__":
    # Set a mock target PIN for testing
    secret_pin = "7392"
    simulate_brute_force(secret_pin)
```

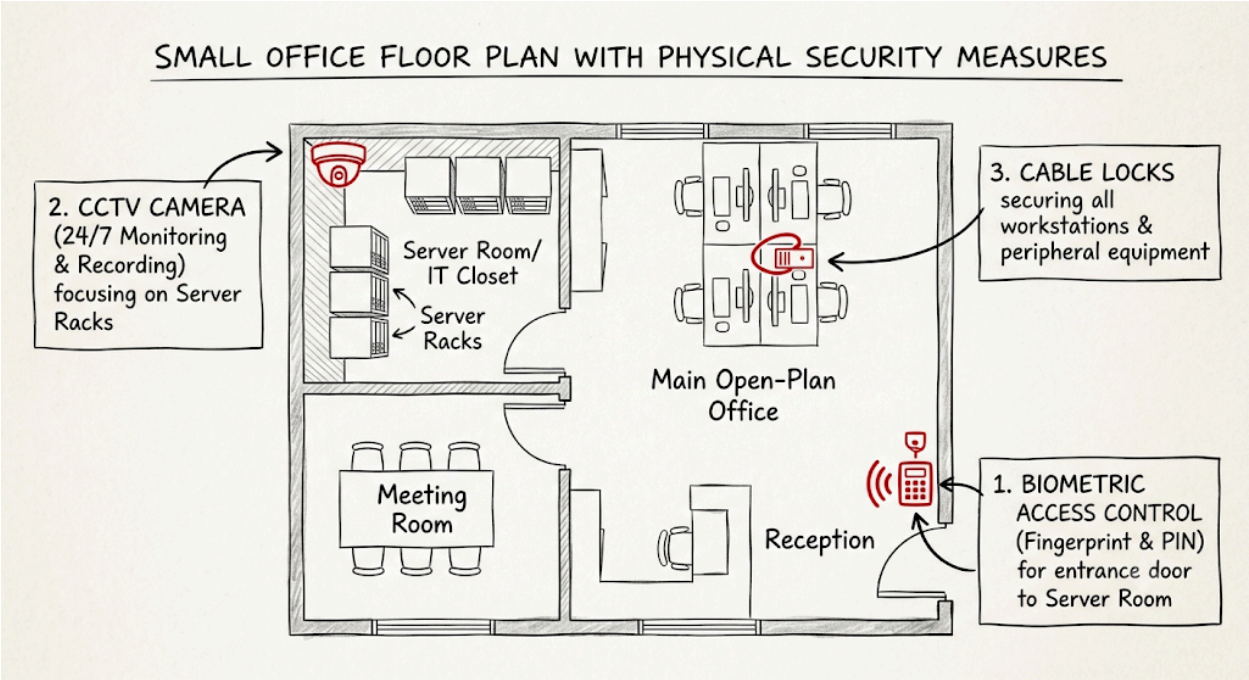
List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

ANS:

Unauthorized Access: Use biometric authentication and mantrap airlock entry portals.
Fire Hazards: Install clean-agent gaseous fire suppression systems (no water damage).
Climate Failures: Deploy redundant N+1 cooling units and real-time thermal monitoring.
Power Outages: Maintain uninterruptible power supply (UPS) banks and backup diesel generators.

Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

ANS:



Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

ANS:

VLAN Segmentation: Isolate the lab network from administrative systems to contain infections.

BIOS Lockdown: Password-protect BIOS and disable booting from external USB or optical media.

Physical Port/Cable Locks: Secure towers and monitors with steel cables and block unused ports.

Disk Rollback Software: Use deep-freeze tools to wipe temporary files and malware upon reboot.

802.1X Port Security: Enforce authentication so only authorized lab hardware can connect to switches.

ID Badge Access & CCTV: Install student ID badge readers at doors and monitor aisles with 24/7 cameras.

Content Filtering & IDS: Block phishing sites and malicious traffic via centralized firewalls.